Class:- S.E.(Computer) **Semester:-**III **Academic Year:-**2026-27

Div :- C **Name of Subject:- Data Structures- PCC-201-COM**

Question Bank of Unit I

Q.1	What is Algorithm ,Flowchart , Data structure
Q.2	Differentiate Data and Information
Q.3	Differentiate Linear Data Structure and Non-linear Data Structure
Q.4	Explain Classification of Data Structure
Q.5	Write a Short Note on- 1. Linear data structures 2. Persistent data structures 3. Non-linear data structures 4.Dynamic data structures 5. Static data structures 6.Ephemeral data structures
Q.6	Explain Asymptotic notation with example 1. Big-O Asymptotic notation 2. Omega Asymptotic notation 3.Theta Asymptotic notation
Q.7	Explain Divide and Conquer with example.
Q.8	Explain Greedy strategy with example.
Q.9	Explain analysis of programming constructs

Q.10	Apply Step count method to evaluate time complexity of the following python code:
	i) function isPrime(n)
	{ for (i = 2; i < n; ++i)
	{ if (n % i == 0)
	return false;
	}
	return true;
	}
	ii) Function f()
	{ ans = 0;
	for (i = n; i >= 1; i = i / 2)
	{ for (j = 1; j <= m; j *= 2)
	{ ans += (i * j);
	}
	}
	print(ans);
	}
	iii) int x = 0;
	for (i = 0; i<N;i++)
	{ for (j =N; j > i ;i--)
	x = x + i + j;
	}
	}

Q.11	Draw flowchart to check whether a given number is a perfect square of an integer. What is the time complexity of your algorithm